A First Course in Abstract Algebra

Part 8: Groups in Topology

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

Contents

41 Simplicial Complexes and Homology Groups 4

Exercise 41.1: Boundary of the specified 2-chain 4

Exercise 41.2: The boundary of a tetrahedron boundary 4

Exercise 41.3: The chain groups of one point 4

Exercise 41.4: Two isolated points 4

Exercise 41.5: A closed line segment 5

Exercise 41.6: True or false: chains, cycles, and boundaries 5

Exercise 41.7: Oriented simplexes in arbitrary dimension 5

Exercise 41.8: Chain, cycle, and boundary groups in general 5

Exercise 41.9: Why the boundary squared is zero 6

Exercise 41.10: Coboundaries in the solid tetrahedron 6

Exercise 41.11: Defining the coboundary and checking its square 6

Exercise 41.12: Cocycles and coboundaries 7

Exercise 41.13: Cohomology of the tetrahedron surface 7

42 Computations of Homology Groups 8

Exercise 42.1: A figure eight 8

Exercise 42.2: Two tangent 2-spheres . . . 8
Exercise 42.3: A sphere and a disjoint annulus . . . 8
Exercise 42.4: An annulus attached along an equator . . . 8
Exercise 42.5: A circle touching a sphere . . . 9
Exercise 42.6: A sphere with one handle . . . 9
Exercise 42.7: True or false: homology invariance . . . 9
Exercise 42.8: Two disjoint torus surfaces . . . 9
Exercise 42.9: Two tori stacked along a common circle . . . 10
Exercise 42.10: A balloon wearing an inner tube . . . 10
Exercise 42.11: A sphere with two handles . . . 10
Exercise 42.12: A sphere with n handles . . . 10
43 More Homology Computations and Applications . . . 11
Exercise 43.1: Two triangulations of a square . . . 11
Exercise 43.2: Euler characteristic and Betti numbers . . . 11
Exercise 43.3: Fixed points on a square and on two disks . . . 11
Exercise 43.4: A sphere touching a Klein bottle . . . 11
Exercise 43.5: Two disjoint Klein bottles . . . 12
Exercise 43.6: True or false: general homology properties . . . 12
Exercise 43.7: Euler characteristic of a surface with handles . . . 12
Exercise 43.8: Triangulating and computing the projective plane . . . 13
Exercise 43.9: A sphere with q cross caps . . . 13
Exercise 43.10: Induced maps of torus coverings . . . 14
Exercise 43.11: Maps from the torus onto its longitude . . . 14
Exercise 43.12: The same maps viewed as torus self-maps . . . 14
Exercise 43.13: Projecting the Klein bottle onto b . . . 15
44 Homological Algebra . . . 16
Exercise 44.1: An exact sequence with two nonzero terms . . . 16
Exercise 44.2: A short exact sequence . . . 16
Exercise 44.3: Three equivalent conditions in an exact sequence . . . 16
Exercise 44.4: When the middle map is an isomorphism . . . 16
Exercise 44.5: The induced-quotient-map principle . . . 16
Exercise 44.6: Relative homology of a torus modulo a . . . 17
Exercise 44.7: The exact sequence of the torus pair . . . 17
Exercise 44.8: Relative homology of a Klein bottle modulo a . . . 17

Exercise 44.9: The exact sequence of the Klein-bottle pair 18
Exercise 44.10: An annulus relative to both boundary circles 18
Exercise 44.11: The exact sequence for the annulus pair 18
Exercise 44.12: The quotient projection is a chain map 18
Exercise 44.13: Well-definedness of the connecting map 19
Exercise 44.14: Consecutive homology-sequence maps compose to zero 19
Exercise 44.15: Exactness of the long homology sequence 19
Exercise 44.16: Chain homotopies induce equal homology maps 19

41 Simplicial Complexes and Homology Groups

We use integer coefficients and ordinary, unreduced homology. Brackets list the vertices of an oriented simplex; exchanging two vertices reverses its sign. The zero group is denoted by 0.

Exercise 41.1: Boundary of the specified 2-chain

Use $[ijk] = P_iP_jP_k$ with the order indicating orientation. Reversing two vertices changes the sign, so the chain in the question becomes

$$c = 2[134] - 4[346] - 3[234] - [146].$$

(a) Apply $\partial[ijk] = [jk] - [ik] + [ij]$ term by term:

$$\begin{aligned}\partial c &= 2([34] - [14] + [13]) - 4([46] - [36] + [34]) \\ &\quad - 3([34] - [24] + [23]) - ([46] - [16] + [14]) \\ &= 2[13] - 3[14] + [16] - 3[23] + 3[24] \\ &\quad - 5[34] + 4[36] - 5[46].\end{aligned}$$

(b) No: distinct oriented edges form a basis of the chain group, and for instance the coefficient of [46] is $-5 \neq 0$. Thus $\partial c \neq 0$. **(c)** Yes: $\partial(\partial c) = 0$ because the boundary of a boundary is zero. This can also be checked by replacing each $[ij]$ by $P_j - P_i$; every vertex coefficient cancels.

Exercise 41.2: The boundary of a tetrahedron boundary

First

$$\partial_3[1234] = [234] - [134] + [124] - [123].$$

Applying ∂_2 gives

$$\begin{aligned}&([34] - [24] + [23]) - ([34] - [14] + [13]) \\ &\quad + ([24] - [14] + [12]) - ([23] - [13] + [12]) = 0.\end{aligned}$$

Every edge occurs twice with opposite sign. Thus $\partial_2\partial_3[1234] = 0$, as required. The signs, rather than an absence of geometric edges, are what cause the cancellation.

Exercise 41.3: The chain groups of one point

There is exactly one vertex and no positive-dimensional simplex. Hence $C_0(P) = \mathbb{Z}P$ and $C_i(P) = 0$ for $i > 0$. Since ∂_0 maps to zero, $Z_0(P) = C_0(P)$. Since $C_1(P) = 0$, $B_0(P) = 0$. Therefore $H_0(P) \cong \mathbb{Z}$. For $i > 0$, all of C_i, Z_i, B_i, H_i are zero. These are ordinary homology groups; reduced H_0 would instead vanish.

Exercise 41.4: Two isolated points

Here $C_0(X) = \mathbb{Z}P \oplus \mathbb{Z}P'$ and $C_i(X) = 0$ for $i > 0$, because no connecting edge is present. Thus $Z_0 = C_0$, $B_0 = 0$, and $H_0 \cong \mathbb{Z}^2$. All positive-dimensional chain, cycle, boundary, and homology groups are zero. The two independent generators of H_0 record the two connected components; the geometric line segment in the surrounding Euclidean space contributes nothing because it is not part of X.

Exercise 41.5: A closed line segment

Let $e = [12]$. Then $C_1 = \mathbb{Z}e$, $C_0 = \mathbb{Z}P_1 \oplus \mathbb{Z}P_2$, and all C_i for $i \geq 2$ vanish. The boundary satisfies $\partial_1(ne) = n(P_2 - P_1)$. Since the vertices are independent, the boundary is zero only if $n = 0$. Thus $Z_1 = 0$; also $B_1 = 0$, so $H_1 = 0$. In dimension zero, $Z_0 = C_0$ and $B_0 = \mathbb{Z}(P_2 - P_1)$. The sum-of-coefficients map $aP_1 + bP_2 \mapsto a + b$ is onto $\mathbb{Z}$ and has kernel B_0, so $H_0 \cong \mathbb{Z}$. In higher dimensions all groups are zero. This describes each of C_i, Z_i, B_i, H_i explicitly.

Exercise 41.6: True or false: chains, cycles, and boundaries

(a) True, because $\partial^2 = 0$.

(b) False. A once-around cycle of a circle cannot bound a 2-chain in that circle, since the circle has no 2-simplexes.

(c) True. One orientation of each n-simplex is a free abelian basis for $C_n(X)$.

(d) True. $B_n(X)$ is a subgroup of the free abelian group $C_n(X)$, so the subgroup theorem for free abelian groups applies.

(e) True for the same reason, with $Z_n(X) \leq C_n(X)$.

(f) True. $H_n = Z_n/B_n$ is a quotient of an abelian group.

(g) False. The boundary of a 3-simplex is a signed sum of four 2-simplexes, not a single 2-simplex.

(h) True. The boundary of a 2-simplex is a signed sum of edges, hence a 1-chain.

(i) True. A 3-cycle has boundary zero, and zero is a valid 2-chain.

(j) True. Quotienting a group by itself gives the one-element group.

Exercise 41.7: Oriented simplexes in arbitrary dimension

(a) An n-simplex is the convex hull of $n+1$ affinely independent vertices $P_0, \ldots, P_n$. An orientation is a choice of ordering modulo even permutations; an odd permutation reverses the orientation (for $n = 0$ there is just its single vertex). **(b)** Its oriented boundary is

$$\partial_n[P_0 \cdots P_n] = \sum_{i=0}^{n} (-1)^i [P_0 \cdots \widehat{P_i} \cdots P_n],$$

where a hat means omission. For example the formula gives $\partial[P_0P_1] = P_1 - P_0$. **(c)** A codimension-one face is obtained by omitting one vertex, with the induced sign $(-1)^i$ in this boundary formula. More generally a k-face is spanned by any $k+1$ of the vertices. In the textbook's oriented-face terminology, choose its orientation so that it appears with positive sign when it is the specified boundary summand.

Exercise 41.8: Chain, cycle, and boundary groups in general

Use the same definitions with an arbitrary integer subscript n: $C_n(X)$ is the free abelian group on a choice of orientation for every n-simplex; opposite orientations have opposite signs. Define

$\partial_n : C_n \to C_{n-1}$ on each basis simplex by the alternating face formula in Exercise 41.7, and extend by integer linearity. Put $C_n = 0$ for $n < 0$ and $\partial_0 = 0$. Then

$$Z_n(X) = \ker \partial_n, \qquad B_n(X) = \operatorname{im} \partial_{n+1}.$$

Thus a cycle is a chain whose boundary is zero, and a boundary is the boundary of a chain one dimension higher. This gives a uniform answer regardless of how large the dimension is.

Exercise 41.9: Why the boundary squared is zero

By linearity it suffices to start with one oriented n-simplex $[P_0 \cdots P_n]$. Every codimension-two face omits two vertices P_i, P_j with $i < j$. There are exactly two ways for this face to occur in the double boundary. Omitting P_i first moves P_j into position $j-1$, giving sign $(-1)^i(-1)^{j-1}$. Omitting P_j first leaves P_i in position i, giving sign $(-1)^j(-1)^i$. These signs are opposites, so the two copies cancel. Every summand is paired this way, proving $\partial_{n-1}\partial_n = 0$ for $n \geq 2$. For $n = 1$ the conclusion follows from $\partial_0 = 0$.

Exercise 41.10: Coboundaries in the solid tetrahedron

Fix increasing vertex order as the positive orientation. The coefficient of a simplex τ in $\delta\sigma$ is the coefficient of σ in $\partial\tau$; this retains the orientation signs in the question. **(a)** Since $\partial[ij] = P_j - P_i$, we have

$$\delta^{(0)}P_1 = -[12] - [13] - [14], \qquad \delta^{(0)}P_4 = [14] + [24] + [34].$$

(b) The oriented edge $P_3P_2 = -[23]$ occurs with coefficient -1 in the boundaries of $[123]$ and $[234]$, and not in the other two faces. Thus

$$\delta^{(1)}(P_3P_2) = -[123] - [234].$$

(c) Since $P_3P_2P_4 = -[234]$ and $[234]$ has coefficient $+1$ in $\partial[1234]$,

$$\delta^{(2)}(P_3P_2P_4) = -[1234].$$

Each negative sign describes the opposite orientation, not an omitted geometric face.

Exercise 41.11: Defining the coboundary and checking its square

(a) For the finite complexes used here identify $C^{(n)}(X)$ with integer coefficient vectors on the oriented n-simplexes. If D_{n+1} is the boundary matrix, define $\delta^{(n)}$ to have matrix D_{n+1}^{T}. Equivalently, sum the oriented cofaces with their incidence signs and extend linearly. **(b)** Since $D_{n+1}D_{n+2} = 0$, transposing gives

$$D_{n+2}^{\mathsf{T}} D_{n+1}^{\mathsf{T}} = 0.$$

This is precisely $\delta^{(n+1)}\delta^{(n)} = 0$. *Scope qualification.* For an arbitrary infinite complex, cochains should be defined as $\operatorname{Hom}(C_n(X), \mathbb{Z})$, allowing arbitrary coefficient functions; they need not equal the finite-support chain group. Define $(\delta\alpha)(c) = \alpha(\partial c)$ there. The same square-zero proof then works without a finiteness assumption. The textbook's identification of chains and cochains is valid for its finite examples.

Exercise 41.12: Cocycles and coboundaries

Define

$$Z^{(n)}(X) = \ker \delta^{(n)}, \qquad B^{(n)}(X) = \operatorname{im} \delta^{(n-1)}.$$

For $n = 0$ set $B^{(0)} = 0$. Kernels and images of homomorphisms are subgroups. If $b = \delta^{(n-1)}c$, then $\delta^{(n)}b = 0$ because $\delta^2 = 0$. Therefore every coboundary is a cocycle and $B^{(n)}(X) \leq Z^{(n)}(X)$, making the next quotient definition possible.

Exercise 41.13: Cohomology of the tetrahedron surface

Define $H^{(n)}(X) = Z^{(n)}(X)/B^{(n)}(X)$. For the surface S there are four vertices, six edges, four faces, and no 3-simplex. In dimension zero $\delta f = 0$ says that f takes the same value at both ends of every edge, so f is constant. Hence $H^{(0)}(S) \cong \mathbb{Z}$. For dimension one, a cocycle has edge values u_{ij} satisfying $u_{jk} - u_{ik} + u_{ij} = 0$ on every triangular face. Put $t_1 = 0$, $t_j = u_{1j}$; the face $[1ij]$ gives $u_{ij} = t_j - t_i$. Thus every 1-cocycle is the coboundary of a vertex function and $H^{(1)}(S) = 0$. In dimension two every cochain is a cocycle, since there are no 3-simplexes. Write e_{ijk} for the cochain dual to a face. Coboundaries of edges give, among others, the relations

$$e_{123} + e_{124} = 0, \quad -e_{123} + e_{134} = 0, \quad e_{123} + e_{234} = 0.$$

The other edge relations follow from these. Thus all four face classes are signed multiples of one unrestricted generator; there is no torsion relation. Consequently

$$H^{(2)}(S) \cong \mathbb{Z}, \qquad H^{(n)}(S) = 0 \ (n > 2).$$

These are cohomology computations from the coboundary maps, rather than an assumption that cohomology always equals homology.

42 Computations of Homology Groups

All homology groups have integer coefficients. Each answer includes H_0: connected spaces contribute one copy of $\mathbb{Z}$ there, whereas disjoint unions contribute one copy per component. All spaces in the exercises are at most two-dimensional, so groups above dimension two vanish.

Exercise 42.1: A figure eight

The two circles meet at exactly one point, so the space is connected and $H_0 \cong \mathbb{Z}$. Triangulate each circle with the shared point as a vertex. A 1-cycle can go independently around either circle, giving two independent cycle generators. There are no 2-simplexes and hence no nonzero boundaries in dimension one. Thus $H_1 \cong \mathbb{Z}^2$, while $H_n = 0$ for $n \geq 2$. This is the wedge of two circles, not the disjoint union of two circles.

Exercise 42.2: Two tangent 2-spheres

The touching point connects the two spheres, so $H_0 \cong \mathbb{Z}$. Each sphere has an independent fundamental 2-cycle, and no 3-chains impose a relation between them. Thus $H_2 \cong \mathbb{Z}^2$. A 1-cycle on either sphere bounds, and passing through the common point does not introduce a new circular passage, so $H_1 = 0$. All H_n for $n \geq 3$ vanish. Algebraically this is the wedge of two 2-spheres, whose positive-dimensional homology is the direct sum of that of the two pieces.

Exercise 42.3: A sphere and a disjoint annulus

There are two components, giving $H_0 \cong \mathbb{Z}^2$. Chains split uniquely between the disjoint components, so each positive-dimensional homology group is the direct sum of those for the pieces. The annulus retracts onto a circle and contributes $\mathbb{Z}$ in dimension one; the sphere contributes $\mathbb{Z}$ in dimension two. Therefore

$$H_1 \cong \mathbb{Z}, \qquad H_2 \cong \mathbb{Z}, \qquad H_n = 0 \ (n \geq 3).$$

Disjointness affects H_0 even though the higher-dimensional contributions can simply be added.

Exercise 42.4: An annulus attached along an equator

Move each point of the annulus radially toward its inner boundary, keeping that boundary and the entire sphere fixed. This deformation stays within the space and retracts the attached annulus onto its inner circle, which already lies on the sphere. Hence the whole space has the homology of the sphere:

$$H_0 \cong \mathbb{Z}, \quad H_1 = 0, \quad H_2 \cong \mathbb{Z}, \quad H_n = 0 \ (n \geq 3).$$

There is no additional dimension-one generator: the annulus's circular generator is the sphere's equator and bounds a hemisphere in the sphere.

Exercise 42.5: A circle touching a sphere

At the one touching point the two components become connected, giving $H_0 \cong \mathbb{Z}$. The circle supplies one 1-cycle that cannot bound in the union, while the sphere supplies one fundamental 2-cycle. Thus

$$H_1 \cong \mathbb{Z}, \qquad H_2 \cong \mathbb{Z}, \qquad H_n = 0 \ (n \geq 3).$$

This is a wedge $S^1 \vee S^2$; the independent positive-dimensional cycles lie in different pieces.

Exercise 42.6: A sphere with one handle

A sphere with one handle is a torus. Cut along a meridian and a longitude to obtain the square model with opposite edges identified. These two loops generate H_1, and the boundary of the square is $a + b - a - b = 0$, so it imposes no nonzero relation in homology. The coherently oriented whole surface is one independent 2-cycle. Consequently

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^2, \quad H_2 \cong \mathbb{Z}, \quad H_n = 0 \ (n \geq 3).$$

Exercise 42.7: True or false: homology invariance

(a) True: homeomorphism invariance is one of the stated basic properties.

(b) False. A point and a closed interval both have $H_0 = \mathbb{Z}$ and all higher groups zero, but are not homeomorphic (they even have different cardinalities).

(c) False. S^n and E^n have different top-dimensional homology for $n > 0$; for $n = 0$, S^0 has two points and E^0 one.

(d) False. The connected circle has $H_1 \cong \mathbb{Z}$.

(e) True. A contractible space has the homology of a point, so its positive-dimensional groups vanish.

(f) False. For $n > 0$, $H_n(S^n) \cong \mathbb{Z}$.

(g) True. E^n is contractible.

(h) False. A torus has $H_1 \cong \mathbb{Z}^2$, but $H_1(S^2) = 0$.

(i) False. A torus has nonzero H_1, H_2, while the disk E^2 is contractible.

(j) True. A sphere with one handle is a standard geometric model of the torus.

Exercise 42.8: Two disjoint torus surfaces

A torus has $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^2$, and $H_2 = \mathbb{Z}$. For a disjoint union the chain complex, and thus its homology, is the direct sum of the two chain complexes. Therefore

$$H_0 \cong \mathbb{Z}^2, \qquad H_1 \cong \mathbb{Z}^4, \qquad H_2 \cong \mathbb{Z}^2, \quad H_n = 0 \ (n \geq 3).$$

There are two independent 2-cycles, because the two oriented torus surfaces are separate.

Exercise 42.9: Two tori stacked along a common circle

The stacked inner-tube surfaces meet along one circle, essential in each torus. The union is connected. Label their common circle b, and their separate meridian cycles a_1, a_2. Cutting and reducing cycles as for a torus leaves three independent 1-generators a_1, a_2, b: the two copies of b are identified. Each whole torus still supplies a separate 2-cycle. No extra 2-cycle arises, because a nonzero multiple of their common circle does not bound a chain on either torus. Hence

$$H_0 \cong \mathbb{Z}, \qquad H_1 \cong \mathbb{Z}^3, \qquad H_2 \cong \mathbb{Z}^2, \quad H_n = 0 \ (n \geq 3).$$

For an algebraic check, the circle-to-pieces map on H_1 sends 1 to $(b, -b)$ in $\mathbb{Z}^2 \oplus \mathbb{Z}^2$. It is primitive and injective, so identifying it removes exactly one rank from dimension one and adds none in dimension two. This uses the intended stacked configuration, whose surfaces touch along a circle.

Exercise 42.10: A balloon wearing an inner tube

The intersection circle is essential in the torus but bounds a hemisphere in the sphere. Thus gluing the sphere to the torus kills the torus's corresponding 1-generator b, leaving its other generator a. There are two independent 2-cycles: the whole sphere and the whole torus. There is no additional one, because the common circle maps nontrivially into the torus's H_1 (its map into the sphere's H_1 is zero). Therefore

$$H_0 \cong \mathbb{Z}, \qquad H_1 \cong \mathbb{Z}, \qquad H_2 \cong \mathbb{Z}^2, \quad H_n = 0 \ (n \geq 3).$$

The gluing calculation can be checked by the map $\mathbb{Z} \to \mathbb{Z}^2 \oplus 0$ sending 1 to the primitive common-circle class: it is injective, and its cokernel is $\mathbb{Z}$.

Exercise 42.11: A sphere with two handles

Cutting along two meridian–longitude pairs gives four 1-generators a_1, b_1, a_2, b_2. The boundary of the remaining polygon traverses each generator once in each direction, so its homological boundary is zero and imposes no relation on these four generators. The sum of the coherently oriented triangles of the closed surface gives one 2-generator, and any 2-cycle must use equal coefficients across adjacent triangles. Hence

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^4, \quad H_2 \cong \mathbb{Z}, \quad H_n = 0 \ (n \geq 3).$$

This surface is connected; it is different from a disjoint pair of tori.

Exercise 42.12: A sphere with n handles

Each handle supplies a meridian and longitude. Cutting along all $2n$ loops gives one polygon whose attaching boundary is

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_n b_n a_n^{-1} b_n^{-1}.$$

In the abelian chain calculation this boundary is zero, since each edge occurs once with each sign. After the standard elimination of interior triangulation edges there is one vertex, $2n$ cycle generators, and one coherently oriented surface generator, with zero boundary maps. Thus for $n \geq 0$,

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^{2n}, \quad H_2 \cong \mathbb{Z}, \quad H_k = 0 \ (k \geq 3).$$

For $n = 0$ this recovers the sphere; for $n = 1$ it recovers the torus. The surface is orientable, so its top-dimensional generator survives over $\mathbb{Z}$.

43 More Homology Computations and Applications

Exercise 43.1: Two triangulations of a square

In the first triangulation there are four corner vertices, four boundary edges and one diagonal, and two triangles. Thus

$$\chi(X) = V - E + F = 4 - 5 + 2 = 1.$$

In the second, the crossing of the diagonals is a vertex, and the horizontal line also introduces a vertex at each midpoint of the left and right sides. Thus there are $4 + 1 + 2 = 7$ vertices. There are six boundary edges and six interior edges (four half-diagonals and two horizontal half-segments), making twelve edges, and six triangles. Hence

$$\chi(X) = 7 - 12 + 6 = 1.$$

Counting intersections and subdivided sides correctly is essential; the two answers agree despite different simplex counts.

Exercise 43.2: Euler characteristic and Betti numbers

The theorem says $\chi = b_0 - b_1 + b_2$ for these two-dimensional spaces, where b_i is the free rank of H_i; torsion contributes no rank. **(a)** The annulus triangulation has five inner and five outer vertices, ten boundary edges and ten other edges, and ten triangles. Thus $\chi = 10 - 20 + 10 = 0$. Its homology has ranks $(1, 1, 0)$, giving $1 - 1 + 0 = 0$. **(b)** In the torus's 3×3 square grid with each small square cut diagonally, opposite sides are identified. This leaves $V = 9$, $E = 27$ (nine horizontal, nine vertical, nine diagonal), and $F = 18$. Thus $\chi = 9 - 27 + 18 = 0$. Its ranks $(1, 2, 1)$ give $1 - 2 + 1 = 0$. **(c)** The same grid with the Klein-bottle edge identifications again gives $V = 9, E = 27, F = 18$, hence $\chi = 0$. Its groups are $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z} \oplus \mathbb{Z}_2$, $H_2 = 0$, so its ranks $(1, 1, 0)$ give $1 - 1 + 0 = 0$. The $\mathbb{Z}_2$ factor does not change the Euler characteristic.

Exercise 43.3: Fixed points on a square and on two disks

For a square the answer is yes: it is homeomorphic to the closed disk E^2, and conjugating a continuous self-map by such a homeomorphism reduces the question to Brouwer's fixed-point theorem. A fixed point of the conjugate map pulls back to one of the original map. For two disjoint disks the answer is no. Choose matching coordinates on the disks and define a map that sends every point of the first to the corresponding point of the second, and conversely. It is continuous on each component, hence on their disjoint union, but has no fixed point because every point changes components. The connected disk hypothesis cannot simply be dropped.

Exercise 43.4: A sphere touching a Klein bottle

The union is connected and is the wedge of a sphere and a Klein bottle at one point. Its positive-dimensional homology is the direct sum of that of the two pieces: cycles on either piece have no new identification except at a vertex. The Klein bottle contributes $\mathbb{Z} \oplus \mathbb{Z}_2$ in dimension one and nothing in dimension two; the sphere contributes $\mathbb{Z}$ in dimension two. Thus

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z} \oplus \mathbb{Z}_2, \quad H_2 \cong \mathbb{Z}, \quad H_n = 0 \ (n \geq 3).$$

Exercise 43.5: Two disjoint Klein bottles

A disjoint union splits its chain complex into the two component chain complexes. Taking direct sums of $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z} \oplus \mathbb{Z}_2$, $H_2 = 0$ for a Klein bottle gives

$$H_0 \cong \mathbb{Z}^2, \qquad H_1 \cong \mathbb{Z}^2 \oplus \mathbb{Z}_2^2, \qquad H_n = 0 \ (n \geq 2).$$

There are two torsion generators, one from each component, and two H_0 generators because the components are disjoint.

Exercise 43.6: True or false: general homology properties

(a) False for ordinary homology: a nonempty contractible space has $H_0 \cong \mathbb{Z}$. Its positive-dimensional groups vanish.

(b) True. A continuous map induces homomorphisms in each dimension; on simplicial chains the construction sends cycles to cycles and boundaries to boundaries.

(c) True. Each is a quotient of abelian groups.

(d) False. A Klein bottle has a $\mathbb{Z}_2$ torsion summand in H_1, which cannot occur in a free abelian group.

(e) True. H_0 is free abelian on the connected components (for a finite triangulation, one copy of $\mathbb{Z}$ per component).

(f) True. If there are no $(n+1)$-simplexes, $B_n = 0$ and $H_n = Z_n \leq C_n$. A subgroup of a free abelian group is free abelian.

(g) True by the definition of the boundary map.

(h) True because applying one more boundary gives zero.

(i) True because $\partial^2 = 0$.

(j) False. A group with torsion, such as H_1 of the Klein bottle, cannot be a subgroup of the torsion-free group C_1. In general homology classes are cosets, not distinguished chains.

Exercise 43.7: Euler characteristic of a surface with handles

For the sphere with n handles, Exercise 42.12 gives ranks $b_0 = 1$, $b_1 = 2n$, $b_2 = 1$, and all others zero. The Euler–Poincaré formula therefore gives

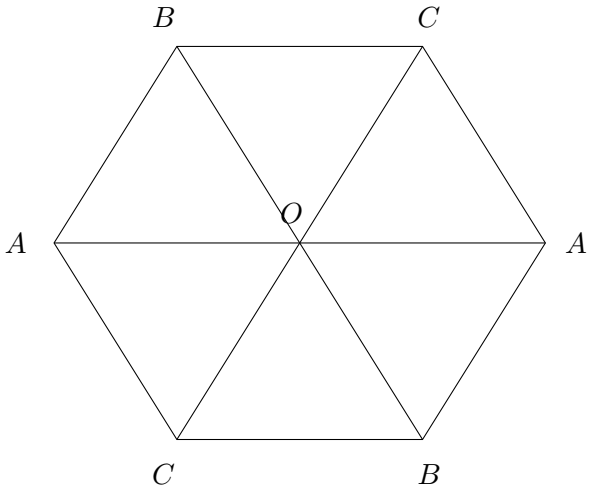
$$\chi = 1 - 2n + 1 = 2 - 2n.$$

This equals 2 for the sphere, 0 for the torus, and decreases by two for every added handle. A polygon cell model also gives the same count: one vertex, $2n$ edges, and one face.

Exercise 43.8: Triangulating and computing the projective plane

Build a genuine triangulation from the disk model.

Replace the circular boundary by a hexagon, keeping the antipodal identification. The paired boundary vertices have labels A, B, C, A, B, C in order; join all six to an interior vertex O.



Before identifying the boundary, barycentrically subdivide each of the six triangles: place a midpoint on each edge, a center in each triangle, and draw the six small triangles corresponding to vertex–edge–triangle flags. Now identify opposite boundary vertices and edge pieces. The subdivision prevents different triangles from having the same vertex set after gluing, so this gives an actual simplicial complex rather than a polygon with folded edges. It has 19 vertices, 54 edges, and 36 triangular faces, with Euler characteristic 1.

Compute the cycles and relations.

Eliminate interior edges using boundaries of the small triangles, just as in the torus calculation. All 1-cycles reduce to multiples of one loop a along the identified semicircles. Traversing the disk boundary goes around a twice in the same direction, so the remaining face boundary is $2a$. Equivalently, the reduced chain calculation has one vertex, one edge a, and one face F, with

$$\partial_1 a = 0, \qquad \partial_2 F = 2a.$$

The elimination consists of choosing and removing interior tree edges and cancelling adjacent face–edge pairs, so it leaves homology unchanged. Thus no nonzero multiple of F is a 2-cycle, while the dimension-one relation is exactly $2a = 0$. The space is connected. Therefore

$$H_0(\mathbb{RP}^2) \cong \mathbb{Z}, \quad H_1(\mathbb{RP}^2) \cong \mathbb{Z}_2, \quad H_n(\mathbb{RP}^2) = 0 \ (n \geq 2).$$

The factor 2 comes from the reversed edge gluing, not from a count of two independent loops.

Exercise 43.9: A sphere with q cross caps

For $q \geq 1$ the polygon model has one vertex, loops $a_1, \ldots, a_q$, and one face whose boundary word is $a_1^2 \cdots a_q^2$. Triangulate the polygon and eliminate interior edges as in Exercise 43.8. The resulting

chain maps are

$$\partial_1 = 0, \qquad \partial_2(1) = 2(a_1 + \cdots + a_q).$$

This map $\mathbb{Z} \to \mathbb{Z}^q$ is injective, so $H_2 = 0$. The vector $u = a_1 + \cdots + a_q$, together with $a_2, \ldots, a_q$, is an integer basis: solve $a_1 = u - a_2 - \cdots - a_q$. In this basis the single relation is $2u = 0$. Hence

$$H_0 \cong \mathbb{Z}, \qquad H_1 \cong \mathbb{Z}^{q-1} \oplus \mathbb{Z}_2, \qquad H_n = 0 \ (n \geq 2).$$

For $q = 1$ this recovers the projective plane; for $q = 2$ it recovers the Klein bottle. With no cross caps, $q = 0$, the space is the sphere instead, with $H_2 = \mathbb{Z}$ and $H_1 = 0$.

Exercise 43.10: Induced maps of torus coverings

In the figure, varying ϕ gives the meridian a, while varying θ gives the longitude b. Use $([a], [b])$ as the ordered basis of $H_1 \cong \mathbb{Z}^2$ and the oriented whole torus $[T]$ as the generator of $H_2 \cong \mathbb{Z}$. In every case $H_0 \cong \mathbb{Z}$ is sent by the identity, since a point maps to a point in a connected space.

(a) For $(\theta, \phi) \mapsto (2\theta, \phi)$, the meridian wraps once and the longitude twice: $[a] \mapsto [a]$, $[b] \mapsto 2[b]$. Thus the H_1 matrix is $\mathrm{diag}(1, 2)$. The whole oriented surface wraps twice, so $[T] \mapsto 2[T]$.

(b) For $(\theta, \phi) \mapsto (\theta, 2\phi)$, $[a] \mapsto 2[a]$, $[b] \mapsto [b]$. The H_1 matrix is $\mathrm{diag}(2, 1)$ and $[T] \mapsto 2[T]$.

(c) For doubling both angles, $[a] \mapsto 2[a]$, $[b] \mapsto 2[b]$. The H_1 matrix is $2I$, and the surface wraps twice in each direction, giving $[T] \mapsto 4[T]$.

The dimension-two multipliers are the determinants of the dimension-one matrices; the positive signs agree with preservation of orientation in each angular direction.

Exercise 43.11: Maps from the torus onto its longitude

The target circle b has $H_0(b) = \mathbb{Z}$, $H_1(b) = \mathbb{Z}[b]$, and $H_2(b) = 0$. Each map sends the H_0 point generator to the point generator, so its H_0 map is the identity. **(a)** For $f(\theta, \phi) = (\theta, 0)$, a meridian becomes a point and the longitude goes around once. Hence $f_*[a] = 0$, $f_*[b] = [b]$, or $(r, s) \mapsto s$ on H_1. **(b)** For $f(\theta, \phi) = (2\theta, 0)$, the same meridian collapses and the longitude goes around twice. Thus $(r, s) \mapsto 2s$ on H_1. In both cases the H_2 map is zero because its target group is zero. Geometrically no two-dimensional surface class can survive a map whose image is a circle.

Exercise 43.12: The same maps viewed as torus self-maps

Now the target has $H_1 = \mathbb{Z}[a] \oplus \mathbb{Z}[b]$ and $H_2 = \mathbb{Z}[T]$, but the maps still factor through the circle b. For part **(a)**, $[a] \mapsto 0$ and $[b] \mapsto [b]$, so the H_1 matrix is $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. For part **(b)**, $[a] \mapsto 0$ and $[b] \mapsto 2[b]$, so it is $\begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$. Both maps induce the identity on H_0. Both induce zero on H_2, even though the target H_2 is nonzero: factorization through b gives a factor through $H_2(b) = 0$. The determinant-zero H_1 matrices provide a consistency check.

Exercise 43.13: Projecting the Klein bottle onto b

The Klein bottle has $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}_2[a] \oplus \mathbb{Z}[b]$, and $H_2 = 0$ in the conventions of Fig. 43.2. The projection to the vertical circle b collapses each horizontal a loop to a point and carries b onto itself once. Therefore the induced maps are

$$H_0 :\ m \mapsto m, \qquad H_1 :\ (\overline{r}, s) \mapsto s, \qquad H_2 :\ 0 \longrightarrow 0.$$

The dimension-one formula is well defined on the order-2 class of a, since that class maps to zero. It also follows algebraically that any homomorphism from its $\mathbb{Z}_2$ summand to the torsion-free group $\mathbb{Z}$ must vanish.

44 Homological Algebra

An exact sequence has image equal to the next kernel at every term. We write 0 for the trivial additive group and use ordinary homology with integer coefficients. A relative group $H_k(X, Y)$ is computed from chains modulo the chains in Y; when Y is nonempty, comparison with X/Y uses reduced homology in dimension zero.

Exercise 44.1: An exact sequence with two nonzero terms

Exactness at A says $\ker f = \operatorname{im}(0 \to A) = 0$, so f is injective. Exactness at B says $\operatorname{im} f = \ker(B \to 0) = B$, so f is surjective. An injective and surjective homomorphism is an isomorphism. Thus the given map itself supplies $A \cong B$.

Exercise 44.2: A short exact sequence

(a) Exactness at C gives $\operatorname{im} j = \ker(C \to 0) = C$, so j is onto. **(b)** Exactness at A gives $\ker i = \operatorname{im}(0 \to A) = 0$, so i is injective and is an isomorphism from A to its image in B. **(c)** Exactness at B gives $\ker j = i[A]$. The first isomorphism theorem, together with surjectivity of j, gives $B/i[A] \cong C$, via $b + i[A] \mapsto j(b)$. Equality of the kernel with $i[A]$ proves that this coset formula is well defined and injective.

Exercise 44.3: Three equivalent conditions in an exact sequence

For the exact sequence $A \xrightarrow{i} B \xrightarrow{j} C \xrightarrow{k} D$, exactness says $\operatorname{im} i = \ker j$ and $\operatorname{im} j = \ker k$. Hence

$$\begin{aligned} i \text{ onto} &\iff \operatorname{im} i = B \iff \ker j = B \\ &\iff j = 0 \iff \operatorname{im} j = 0 \iff \ker k = 0 \iff k \text{ injective.} \end{aligned}$$

The middle condition $j = 0$ means it maps every element of B to the zero element of C. This proves all three requested conditions equivalent.

Exercise 44.4: When the middle map is an isomorphism

For $A \xrightarrow{g} B \xrightarrow{h} C \xrightarrow{i} D \xrightarrow{j} E \xrightarrow{k} F$, exactness yields

$$\operatorname{im} h = \ker i, \quad \operatorname{im} i = \ker j, \quad \operatorname{im} g = \ker h, \quad \operatorname{im} j = \ker k.$$

Thus $h = 0$ is equivalent to injectivity of i, and $j = 0$ is equivalent to surjectivity of i. Together these say exactly that i is an isomorphism $C \to D$, proving (a) $\iff$ (b). Also g is onto exactly when $\ker h = B$, namely when $h = 0$; and k is injective exactly when $\operatorname{im} j = 0$, namely when $j = 0$. This proves (a) $\iff$ (c), completing the equivalence.

Exercise 44.5: The induced-quotient-map principle

Exercise 14.39 gives the criterion: a homomorphism $f : G \to G'$ induces $G/H \to G'/H'$ by $gH \mapsto f(g)H'$ if $f(H) \subseteq H'$. For Theorem 44.4, the chain-map equation first shows $f_k(Z_k(A)) \subseteq Z_k(A')$, since

$\partial' f_k z = f_{k-1}\partial z = 0$. It also shows $f_k(B_k(A)) \subseteq B_k(A')$, since $f_k(\partial c) = \partial'(f_{k+1}c)$. Apply the quotient-map criterion to Z_k/B_k to obtain the induced homology map. For Theorem 44.7, $\partial_k(A'_k) \subseteq A'_{k-1}$ by the subcomplex assumption. Apply the same criterion with $G = A_k$, $H = A'_k$, $G' = A_{k-1}$, $H' = A'_{k-1}$, and $f = \partial_k$. This gives the well-defined quotient boundary $c + A'_k \mapsto \partial c + A'_{k-1}$. Its square is zero because $\partial^2 = 0$. Thus both theorems use precisely the same representative-independence mechanism as Exercise 14.39.

Exercise 44.6: Relative homology of a torus modulo a

After eliminating interior triangulation edges, use the torus chain model with one vertex, edges a, b, and one face F. Its boundaries are $\partial_1 = 0$ and $\partial_2 F = a + b - a - b = 0$. The circle subcomplex a contains the vertex and edge a. In the relative chain complex these are set to zero, leaving

$$C_2(X, a) = \mathbb{Z}F, \qquad C_1(X, a) = \mathbb{Z}b, \qquad C_0(X, a) = 0,$$

with zero boundary maps. Hence

$$H_2(X, a) \cong \mathbb{Z}, \quad H_1(X, a) \cong \mathbb{Z}, \quad H_0(X, a) = 0,$$

and higher groups vanish. Collapsing a gives a pinched torus with the positive-dimensional homology of $S^2 \vee S^1$. There is a necessary qualification to the printed parenthesis: relative homology corresponds to *reduced* homology of the quotient. The quotient's ordinary H_0 is $\mathbb{Z}$, whereas the relative H_0 computed here is zero.

Exercise 44.7: The exact sequence of the torus pair

Use $H_1(X) = \mathbb{Z}[a] \oplus \mathbb{Z}[b]$ and choose matching orientations for the whole-surface generators. The nonzero part of the exact sequence is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{1} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{u} \mathbb{Z}^2$$
$$\xrightarrow{v} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{1} \mathbb{Z} \longrightarrow 0,$$

where the groups in order are $H_2(X)$, $H_2(X, a)$, $H_1(a)$, $H_1(X)$, $H_1(X, a)$, $H_0(a)$, $H_0(X)$. Here $u(t) = (t, 0)$ and $v(r, s) = s$. The first map carries the whole torus to the same relative 2-cycle. The first connecting map is zero because that 2-cycle has zero boundary. The second connecting map is zero because b already has zero boundary. To check exactness directly: the identity map has full image, equal to the next zero map's kernel; the zero image equals $\ker u = 0$; $\operatorname{im} u = \mathbb{Z} \times 0 = \ker v$; v is onto, matching the next zero map's full kernel; and the final identity has zero kernel. This checks every position.

Exercise 44.8: Relative homology of a Klein bottle modulo a

For the Klein-bottle polygon use one vertex, edges a, b, and a face F with $\partial_2 F = 2a$ and $\partial_1 = 0$. Set the vertex and the edge a to zero in the relative complex. Then $\partial_2 F$ also becomes zero, leaving one free generator in dimensions two and one and no dimension-zero generator. Thus

$$H_2(X, a) \cong \mathbb{Z}, \qquad H_1(X, a) \cong \mathbb{Z}, \qquad H_0(X, a) = 0,$$

with higher groups zero. This explains why collapsing the torsion loop creates a top-dimensional cycle: its original boundary was twice the collapsed edge. As in Exercise 44.6, the connected pinched quotient has ordinary $H_0 = \mathbb{Z}$; its *reduced* homology agrees with these relative groups.

Exercise 44.9: The exact sequence of the Klein-bottle pair

Here $H_2(X) = 0$, $H_1(X) = \mathbb{Z}_2[a] \oplus \mathbb{Z}[b]$, and $H_0(X) = \mathbb{Z}$. The relevant sequence is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{u} \mathbb{Z}_2 \oplus \mathbb{Z}$$
$$\xrightarrow{v} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{1} \mathbb{Z} \longrightarrow 0.$$

The first $\mathbb{Z}$ is $H_2(X, a)$; the next is $H_1(a)$; the later three $\mathbb{Z}$ groups are $H_1(X, a)$, $H_0(a)$, $H_0(X)$. The connecting map sends F to its boundary $2a$, so it is multiplication by 2. Also $u(t) = (\overline{t}, 0)$ and $v(\overline{r}, s) = s$. Check exactness: multiplication by 2 is injective, $\ker u = 2\mathbb{Z}$, $\operatorname{im} u = \mathbb{Z}_2 \oplus 0 = \ker v$, v is onto, the following map is zero, and the final identity has zero kernel. These identities verify every term, including the torsion contribution.

Exercise 44.10: An annulus relative to both boundary circles

Let the boundary circles have the same angular orientation, and let Y be their union. A coherently oriented sum of the annulus's triangles has boundary equal to the difference of those two circles, which becomes zero modulo Y. It gives one relative 2-cycle F. A radial path e from one boundary component to the other has boundary in Y, so it gives one relative 1-cycle. Cutting the annulus along this path produces a disk; eliminating interior triangle edges shows that F and e exhaust the independent relative cycles. In a reduced relative chain model, there is one generator in degrees two and one, no generator in degree zero, and zero boundary maps. Hence

$$H_2(X, Y) \cong \mathbb{Z}, \qquad H_1(X, Y) \cong \mathbb{Z}, \qquad H_0(X, Y) = 0,$$

and higher groups vanish. The relative 1-generator is a radial path, not the circular loop, since that loop is homologous to a boundary circle and is zero in the relative group.

Exercise 44.11: The exact sequence for the annulus pair

Use the same angular orientation on the two boundary circles. Then inclusion maps $H_1(Y) = \mathbb{Z}^2 \to H_1(X) = \mathbb{Z}$ by $s(r, t) = r + t$. The analogous H_0 map is also s. Choose the orientation of the relative fundamental 2-chain and radial 1-chain so that their connecting maps are $d(k) = (k, -k)$. The nonzero part of the sequence is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d} \mathbb{Z}^2 \xrightarrow{s} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{d} \mathbb{Z}^2 \xrightarrow{s} \mathbb{Z} \longrightarrow 0.$$

The groups are respectively $H_2(X, Y)$, $H_1(Y)$, $H_1(X)$, $H_1(X, Y)$, $H_0(Y)$, $H_0(X)$; $H_2(X)$ and $H_0(X, Y)$ are zero. Directly, d is injective, $\operatorname{im} d = \{(k, -k)\} = \ker s$, and s is onto. These facts check exactness at both pairs of $\mathbb{Z}^2$ and $\mathbb{Z}$ terms. At the middle relative $\mathbb{Z}$, the incoming zero map has image zero, equal to the next d's kernel; at the preceding $\mathbb{Z}$, the onto map s fills the kernel of the zero map.

Exercise 44.12: The quotient projection is a chain map

Let $j_k(c) = c + A'_k$. By the definition of the quotient boundary,

$$\overline{\partial}_k(j_k(c)) = \overline{\partial}_k(c + A'_k) = \partial_k c + A'_{k-1} = j_{k-1}(\partial_k c).$$

Thus $\overline{\partial}_k j_k = j_{k-1}\partial_k$ for every k. This proves the squares commute, so the quotient projections constitute a chain map.

Exercise 44.13: Well-definedness of the connecting map

A relative homology class is represented by $c \in A_k$ with $\partial c \in A'_{k-1}$. Since $\partial^2 c = 0$, this boundary is a cycle of A', so $[\partial c] \in H_{k-1}(A')$ is meaningful. Two choices representing the same relative homology class differ by

$$c' - c = \partial d + a', \qquad d \in A_{k+1}, \quad a' \in A'_k.$$

Indeed equality modulo relative boundaries is precisely this condition, including the possible change of coset representative in A_k/A'_k. Taking boundaries gives $\partial c' - \partial c = \partial a'$, a boundary *inside* A'. Thus $[\partial c'] = [\partial c]$ there, proving independence of both choices. Finally $\partial(c_1 + c_2) = \partial c_1 + \partial c_2$, so the rule $[c] \mapsto [\partial c]$ preserves addition. It is therefore a well-defined homomorphism.

Exercise 44.14: Consecutive homology-sequence maps compose to zero

There are three kinds of consecutive composites to check. First $j_* i_* [z'] = [z' + A'_k] = 0$, since a chain in the subcomplex is zero in the quotient. Second $\partial_* j_* [z] = [\partial z] = 0$, because a homology representative z in A is a cycle. Third $i_* \partial_* [c] = [\partial c] = 0$ in $H_{k-1}(A)$, because ∂c is a boundary in A (even though it need not be a boundary in A'). Thus every consecutive composite is zero. All groups are abelian and all maps are homomorphisms, so the long sequence is indeed a chain complex.

Exercise 44.15: Exactness of the long homology sequence

Use $[z]$ for an absolute homology class, and use $[c]_{\mathrm{rel}}$ for a relative class, with $\partial c \in A'$.

(a) If a class comes from $H_k(A')$, its chain is zero in A_k/A'_k. Thus $\operatorname{im} i_* \subseteq \ker j_*$.

(b) If $j_*[z] = 0$, the cycle $z \in A_k$ satisfies $z = \partial d + a'$ with $d \in A_{k+1}$ and $a' \in A'_k$. Taking boundaries gives $\partial a' = 0$. Hence $[z] = [a'] = i_*[a']$ in $H_k(A)$, so $\ker j_* \subseteq \operatorname{im} i_*$.

(c) For an absolute cycle z, $\partial_* j_* [z] = [\partial z] = 0$. Thus $\operatorname{im} j_* \subseteq \ker \partial_*$.

(d) If $\partial_* [c]_{\mathrm{rel}} = 0$, there is $a' \in A'_k$ with $\partial c = \partial a'$. Then $z = c - a'$ is a cycle in A, and $j_*[z] = [c]_{\mathrm{rel}}$ since a' vanishes in the quotient. Therefore $\ker \partial_* \subseteq \operatorname{im} j_*$.

(e) The class $\partial_* [c]_{\mathrm{rel}} = [\partial c]$ maps to zero in $H_{k-1}(A)$ because it bounds c there. Thus $\operatorname{im} \partial_* \subseteq \ker i_*$.

(f) If $[a'] \in H_{k-1}(A')$ maps to zero in $H_{k-1}(A)$, then $a' = \partial c$ for some $c \in A_k$. Its boundary is in A', so it represents a relative cycle, and $\partial_* [c]_{\mathrm{rel}} = [a']$. Thus $\ker i_* \subseteq \operatorname{im} \partial_*$.

The six inclusions pair into equality at all three types of term, in every dimension. This proves exactness of the entire sequence.

Exercise 44.16: Chain homotopies induce equal homology maps

Let $z \in Z_k(A)$ represent a homology class. Because $\partial z = 0$, the given homotopy identity reduces to

$$f_k(z) - g_k(z) = \partial'_{k+1}(D_k z) + D_{k-1}(0) = \partial'_{k+1}(D_k z).$$

Thus the two image cycles differ by a boundary in A', so they represent the same class in $H_k(A')$. Since every homology class has a cycle representative, $f_{*k} = g_{*k}$ for every k. The chain-map hypotheses

ensure both induced maps are defined; the homotopy supplies the boundary witnessing equality on each class.